

Ceph Cheat Sheet

Ceph is a unified, distributed storage system providing **Object, Block, and File storage** from a single cluster.

🏗️ Core Architecture & Components

- * **RADOS (Reliable Autonomic Distributed Object Store)**: The foundation of all Ceph clusters.
- * **Ceph OSD (Object Storage Daemon)**: Stores data, handles replication, recovery, and rebalancing. (1 per disk)
- * **Ceph Monitor (MON)**: Maintains cluster maps (Mon, OSD, PG, CRUSH) and ensures high availability via Paxos.
- * **Ceph Manager (MGR)**: Provides monitoring, orchestration, and dashboard services.
- * **Ceph Metadata Server (MDS)**: Stores metadata for CephFS (not needed for Block/Object).
- * **RADOS Gateway (RGW)**: Object storage interface (S3/Swift compatible).
- * **CRUSH Algorithm**: Calculates data placement without a central lookup table.

🛠️ Essential CLI Commands

🔗 Cluster Health & Status

Command	Description
:---	:---
`ceph health [detail]`	Check cluster health status
`ceph -s` / `ceph status`	Show cluster status summary
`ceph -w`	Watch cluster events in real-time
`ceph df`	Show cluster storage usage
`ceph osd tree`	View OSD hierarchy and status

📁 Pool Management

Command	Description
:---	:---
`ceph osd lspools`	List all pools
`ceph osd pool create <name> <pg_num>`	Create a new pool
`ceph osd pool delete <name> <name> --yes-i-really-really-mean-it`	Delete a pool
`ceph osd pool rename <old> <new>`	Rename a pool
`ceph osd pool set <name> size <num>`	Set replica count

🛠️ OSD Management

Command	Description
:---	:---
`ceph osd stat`	Show OSD status summary
`ceph osd out <id>`	Mark OSD as out (starts rebalancing)
`ceph osd in <id>`	Mark OSD as in
`ceph osd down <id>`	Force OSD status to down
`ceph osd df`	Show utilization per OSD

📁 Storage Interfaces

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*    **Block (RBD):** `rbd create <img_name> --size <MB>`, `rbd ls`, `rbd
map <img_name>`
*    **File (CephFS):** `ceph fs ls`, `ceph fs status`, `mount -t ceph
<mon_ip>:/ <mnt_pt>`
*    **Object (RGW):** Managed via `radosgw-admin` (e.g., `radosgw-admin
user create`)
```

🔍 Troubleshooting & Monitoring

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*    **Logs:** Usually in `/var/log/ceph/`
*    **Check Auth:** `ceph auth list`
*    **PG Status:** `ceph pg stat`, `ceph pg dump`
*    **Check Versions:** `ceph versions`
*    **Monitor Quorum:** `ceph mon stat`
```